

Three hundred ways to assess Europe's surface waters: An almost complete overview of biological methods to implement the Water Framework Directive

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ABSTRACT

According to the Water Framework Directive (WFD), the status of European surface waters is assessed using aquatic organism groups. Here we present an overview of 297 assessment methods, based on a questionnaire survey addressing authorities in all countries implementing the WFD. Twenty-eight countries reported on methods applied to rivers (30%), coastal waters (26%), lakes (25%) and transitional waters (19%). More than half of the methods are based on macroscopic plants (28%) or benthic invertebrates (26%); in addition, phytoplankton (21%), fish (15%) and phytobenthos (10%) were assessed. Countries of Central and Western Europe had developed almost all methods required for the WFD implementation. Two main sampling strategies were discernable: small-scale sampling of the taxonomically diverse groups of benthic invertebrates and phytobenthos that demand elaborate processing, versus large-scale sampling of vast, species-poor plant stands or the mobile fish fauna. About three-quarters of methods identified organisms to species-level while in particular phytoplankton-based methods used class- or phylum-level, or included no taxonomic information. Out of nine metric types distinguished, river methods used more sensitivity and trait metrics while for other water categories abundance metrics prevailed. Fish-based methods showed the highest number of metrics. Fifty-six percent of methods focussed on the detection of eutrophication and organic pollution, with shares decreasing from autotrophic to heterotrophic organism groups: phytoplankton > phytobenthos > macroscopic plants > benthic invertebrates > fish fauna. The order was almost reverse for hydrological or morphological deterioration: fish fauna and macroscopic plants > benthic invertebrates > phytoplankton > phytobenthos. These pressures were mainly assessed by methods applied to rivers and transitional waters. The pressure-impact relationship of about one-third of methods was not tested empirically with methods for transitional waters being the least validated. The strength of relationships differed significantly between organism groups and water categories. The correlation coefficients generally covered a broad range (<0.4 to >0.8), but on average with the pattern: phytoplankton > macroscopic plants > benthic invertebrates > phytobenthos and fish fauna. In terms of water categories the following order resulted: coastal waters > lakes > transitional waters > rivers. Status boundaries were mostly defined using statistical approaches. We advocate better reflection of the necessary sampling effort and precision, full validations of pressure-impact relationships and an implementation of more ecological components into classification.

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1. Introduction

Aquatic bioindicators are organisms accumulating toxic substances or responding to environmental stress, such as pollution,

nutrient enrichment, habitat loss or overexploitation (Adams, 2002). Bioindication requires standardized procedures including field sampling, sample processing and identification of collected organisms. The gathered information on occurring taxa is then usually summarised in biological metrics (Karr and Chu, 1999), which are compared to standards or reference values and classified into quality classes. Each of these steps could be done in countless ways: sampling can be performed with different equipment, identification of organisms can be to different taxonomic levels and based on

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different keys, and there is a huge number of assessment metrics and procedures for class boundary setting. In Europe, bioassessment methods differ also geographically, as organism response to stress may vary by region and ecosystem type, different species may occur, relevant stressors may differ and applicable taxonomic resolution may vary with the knowledge of the regional fauna and flora.

Over the last ten years biomonitoring of European aquatic ecosystems changed substantially. The development was driven by the EU Water Framework Directive 2000/60/EC (WFD), which required assessment methods for different ecosystem types (“water categories”: rivers, lakes, transitional waters, coastal waters) and different organism groups (“biological quality elements” = BQEs: phytoplankton, aquatic flora, benthic invertebrates, fish). The WFD has changed management objectives from merely pollution control to ensuring ecosystem integrity (Borja et al., 2008). Deterioration and improvement of “ecological status” is defined by the response of the biota, rather than by changes in environmental parameters. This response must be investigated at the level of the “water body” (e.g. a river stretch, a lake or a part of a coastal water), which represents the classification and management unit of the WFD. Water bodies of the same category are grouped into “water body types”, for each of which undisturbed reference states are defined (i.e. the taxa expected in the absence of human pressures). In biological assessment, the observed condition is compared against the reference status with the result given in five classes: high status (no differences to reference conditions), good status (slight differences), moderate status (moderate differences), poor and bad status (major differences). Good ecological status represents the target value that all surface water bodies have to achieve in the near future. While the WFD indicates what characteristics of the BQEs should be assessed (e.g. “abundance”, “community composition”) it does not specify, however, which indices or metrics of these various elements should be used (Hering et al., 2010). This decision was left to the EU Member States.

Just developing methods for the different combinations of BQEs and water categories would have resulted in about 20 methods. However, many countries preferred developing country-specific methods, either to continue using existing time series by adapting their national methods to the WFD, or to regard for the specific ecoregional and biogeographic situation; therefore, a multitude of methods resulted instead of a handful of methods applicable Europe-wide (e.g. Birk and Schmedtje, 2005; Borja et al., 2009).

Here we present for the first time an almost complete overview of bioassessment methods for European aquatic ecosystems which were adapted or newly developed for the WFD. We search for commonalities and differences within the full range of methods, focussing on the implementation of scientific concepts and standards of aquatic bioassessment guided by six questions:

- (a) Do the bioassessment methods cover all BQEs and water categories across Europe? Ideally, the same BQEs should be applied in all EU Member States allowing for an integrated ecological evaluation.
- (b) Is the effort to acquire the biological data sufficient and feasible to fulfil the assessment objectives? Sampling procedure greatly influences the results of bioassessment. Thus, sampling must be sufficiently precise and representative of the relevant BQE characteristics in time and space (de Jonge et al., 2006). At the same time, sampling should not be overcomplicated to save costs.
- (c) Is the taxonomic resolution appropriate for the assessment method? Taxonomic composition represents a key parameter in the assessment of ecological status. The degree of taxonomic resolution is imperative for ecological precision, available expertise and associated costs.

- (d) Is the selection of biological assessment metrics well-balanced? Biological metrics should reflect different types of responses to human disturbances and different levels of ecological hierarchy from individuals to landscape (Karr and Chu, 1999).
- (e) Is the methods' response to anthropogenic pressure known and quantified? Ecological classification should reflect the degree of human influence on the aquatic ecosystem (Borja et al., 2011).
- (f) Does the quality class boundary setting follow an ecological rationale? Boundaries should represent tipping-points in quality defined by thresholds relevant to the ecosystem (Groffman et al., 2006).

2. Material and methods

2.1. Data collection

Data on national assessment methods were collated using a questionnaire. This process was part of the official reporting procedure of the European intercalibration exercise to gain full descriptions of biological assessment methods used in national WFD monitoring programmes (European Commission, 2010). We prepared a questionnaire following the principles described by Oppenheim (1998) and Noelle-Neumann and Petersen (2005), comprising 66 questions that covered the topics general information, data acquisition and data evaluation (see [supplementary data](#)). Addressees included coordinators at national water administrations, members of the WFD Working Group on Ecological Status (ECOSTAT) and leading scientists in the European intercalibration exercise. Expert selection ensured a complete coverage of countries implementing the WFD and experts for each water category and BQE in the individual countries. We asked the recipients to forward the questionnaire to other experts, if necessary.

Data from questionnaires returned until November 2010 were analysed in this publication. The full data were entered into an online database (Birk et al., 2010) accessible at <http://www.wiser.eu/programme-and-results/data-and-guidelines/method-database/>. To complete data gaps or clarify ambiguous information on specific aspects of reported methods, we consulted the scientific literature cited in the questionnaires.

2.2. Data analysis

We described the frequency distributions of selected parameters using pie charts, and compared the parameter values among BQEs and water categories by with the Kruskal-Wallis H-Test. For this we defined analytical units (AU) by comparing the methods between water categories, BQE or pressure-responses depending on the research question. A method that is used in coastal and transitional waters, for instance, was assigned to two AUs. For some analyses we aggregated the BQEs freshwater macrophytes, marine angiosperms and macroalgae into “macroscopic plants”. Together with freshwater phytobenthos these BQEs represent the “benthic flora”.

To analyse the categories of sampling and biological metric selection we applied Multiple Correspondence Analysis (MCA). Homogeneity in the dataset and the absence of rare categories were checked in order to avoid bias (Escofier and Pagès, 1990). For the multivariate analyses data were analysed as complete disjunctive tables. Since the data basis comprised some incomplete information, the number of methods used in the analyses slightly differed with research question (Table 1).

2.2.1. Coverage of aquatic bioassessment

To summarise the outcomes of the questionnaire survey we quantified the total number of biological assessment methods, and described the relative shares of BQEs and water categories covered

Table 1
Number of national methods (N) and analytical units (AU) used in the analyses.

Analysed method components	N	AU
Availability of bioassessment methods		
General overview	297	324 ^b , 300 ^c
Methods' completeness	290 ^a	290
Sampling		
Sampling season	279	308 ^d
Sample size	196	208 ^c
Sampling site selection	278	305 ^d
Taxonomic resolution	288	288
Biological metric selection	290 ^a	1170 ^e
Pressure-impact relationship		
Scope of detected pressures	290	495 ^b , 458 ^c
Tested relationship	269	295 ^b , 282 ^c
Strength of relationship	110	119 ^b , 115 ^c
Number of observations to validate relationship	131	139 ^b , 134 ^c
Quality class boundary setting	259	259

^a Including only methods for which biological assessment metrics were specified.

^b Number of AU among water categories.

^c Number of AU among biological quality elements.

^d "Active individuals" (i.e. analytical units) in Multiple Correspondence Analysis.

^e Biological metrics as "active individuals" in Multiple Correspondence Analysis.

Table 2
Categories of the sampling criteria.

Criterion	Categories
Sample size	Small surface area (<20 m ²) Medium surface area (20–100 m ²) Large surface area (>100 m ²) Entire water body
Sampling season	Spring (mid-March to mid-June) Summer (mid-June to mid-September) Autumn (mid-September to end of November) Winter (December to mid-March) Number of sampled seasons
Sample site selection	Expert-based Random Stratified

by the methods. Furthermore, we evaluated if all BQEs and water categories across Europe were covered by this survey. In theory, a country should have adopted assessment methods for each BQE defined for each water category located within its territory. We related the number of methods that were actually described to this theoretical value to gain the percent of method coverage for each country.

2.2.2. Sampling

In order to investigate the sampling strategies of aquatic bioassessment methods in Europe we analysed the information about sample size, sampling season and site selection. The data provided in the questionnaires were categorised (Table 2) and tested for differences in sample sizes between BQEs and water categories. For the criteria of sampling season and site selection we applied MCA. Aquatic flora was separated into macroscopic plants and phytobenthos as we expected systematic sampling differences between these elements. Phytoplankton methods were excluded from the analysis of sample size as it was not possible to categorise the sampled volumes using the available data.

2.2.3. Taxonomic resolution

By reviewing the list of assessment metrics used by each method we identified which taxonomic level was required to allow for calculating all metrics applied by the method. If, for example, (1) number of families and (2) total abundance of sampled organisms

were used in a multimetric scheme, family-level identification was selected as the lowest taxonomic unit required by the method.

2.2.4. Biological metric selection

Based on existing metric classifications (Karr and Chu, 1999; Hering et al., 2006) we grouped metrics into the following types: (1) metrics that did not account for ecological characteristics but taxonomy ("taxonomy-based metrics"), (2) metrics that account for ecological characteristics ("autecology-based metrics") and (3) non-biotic metrics (e.g. spatial extent of suitable habitats).

The first category ("taxonomy-based metrics") were further subdivided as:

- richness metrics, i.e. number of taxa of a certain organism group (e.g. total taxa richness, richness of Ephemeroptera, Trichoptera and Plecoptera taxa) and proportional richness metrics,
- metrics of abundance and productivity (including age- and size-structure), also expressed as proportion of a total,
- diversity metrics, i.e. a combination of richness and abundance metrics, and
- methods based on taxonomic assemblage comparisons between sites (e.g. multivariate approaches).

The second category ("autecology-based metrics") included metrics in which taxa were characterised by

- sensitivity to disturbance (whereas disturbance could be nutrient enrichment, organic pollution, acidification, etc.),
- autecological traits,
- individual condition (e.g. individual health, physiological characteristics)
- native vs. non-native (alien) taxa.

2.2.5. Pressure-impact relationship

We evaluated the questionnaire replies on the pressures detected by each method, using four categories of pressure: eutrophication/organic pollution, hydrology/morphology (i.e. aquatic habitat deterioration, flow modification, hydromorphological degradation, riparian habitat alteration), other water quality aspects (i.e. acidification, heavy metals, pollution by organic compounds such as DDT, PCB) and unspecific pressure (i.e. general degradation). We checked if the pressure-impact relationship was tested for method development. We reviewed the strength of the relationships (expressed as correlation coefficients) and the number of observations used to empirically validate the relationships. If methods were tested against the same stressor at different water body types we averaged the correlation coefficients per pressure category. Responses to different pressures were kept separately in the analysis.

2.2.6. Quality class boundary setting

We specified five approaches of boundary setting grouped into ecological, statistical and expert-based types of classifications (Table 3). We assigned one classification type to each method according to the approaches specified by the respondent. For multiple statements we followed the hierarchy of setting options established by the European Commission (2003, 2010): If among others ecological approaches were used, we classified the assessment method accordingly. Any use of statistical setting without ecological approaches resulted in a classification as statistical type. Expert-based setting was only assigned if stated by the respondent without mentioning ecological or statistical approaches.

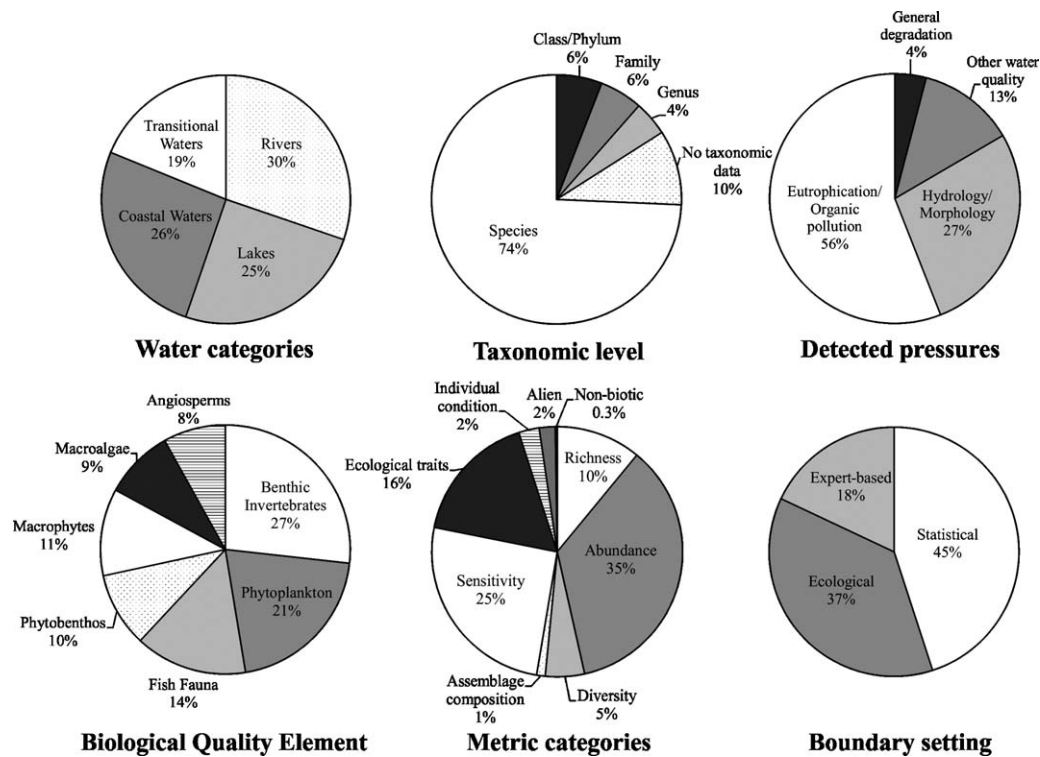


Fig. 1. Relative frequencies of selected features of aquatic assessment methods in Europe.

Table 3

Types of ecological status class boundary setting.

Type of boundary setting	Approaches specified in the questionnaire
Ecological	Using discontinuities in the relationship of anthropogenic pressure and the biological response Using biological metrics that respond in different ways to the influence of the pressure (i.e. "paired" or "crossover" metrics)
Statistical	High-good boundary derived from metric variability at near-natural reference sites Equidistant division of the existing gradient of ecological quality
Expert-based	Class boundaries calibrated against pre-classified sampling sites

3. Results

3.1. Coverage of aquatic bioassessment

A total of 28 European countries reported on 297 biological assessment methods (see [supplementary data](#)). River methods comprised 30% of all methods reported; lake and coastal assessment methods accounted for about 25% each. Transitional waters were least represented (19%) (Fig. 1). Benthic invertebrates were the most prevalent biological group used in aquatic bioassessment followed by macroscopic plants and phytoplankton. Methods evaluating fish fauna and phytobenthos were comparatively rare (Fig. 1), as monitoring of these biological groups is not required for all water categories.

The overview covered 2/3 (66%) of the theoretical number of assessment methods per European Member State (Fig. 2). Methods monitoring invertebrates in rivers and phytoplankton in lakes were most widely adopted (90% and 82%, respectively), and the least adopted methods were those using phytoplankton in rivers and phytobenthos in lakes (both 21%).

3.2. Sampling

Benthic invertebrates and phytobenthos were mainly sampled from small areas of less than 20 m² (Fig. 3). Conversely, macroscopic plants and fish were evaluated based on sampled areas of several 100 m², or even with sampling the whole water body. Phytoplankton was usually sampled in all seasons, while macrophytes were most frequently sampled in summer (Fig. 3). MCA results showed that these findings were independent from the water category. The number of sampling seasons specified by the assessment methods was significantly different between biological elements (H-Test, $p < 0.001$). Sampling for phytoplankton covered most seasons (mean number of 2.7), while macroscopic plants were only surveyed at an average of 1.5 seasons. Expert knowledge was most commonly used to choose sampling sites (78% of assessment methods).

3.3. Taxonomic resolution

Almost three quarters of the methods used species-level data for metric calculation (Fig. 1). Genus or family level information was mainly used by river methods evaluating the benthic invertebrate communities. Class or phylum data was used in phytoplankton assessment. Only 10% of methods were based on non-taxonomic data, e.g. total biomass or abundance, or morphological and physiological features from monospecific angiosperm stands.

3.4. Biological metric selection

Metrics included in the European biomonitoring methods almost equally covered taxonomy-based (about 53% of metrics) and autecology-based (47%) metrics, while non-biotic metrics were rarely used (<0.5%) (Fig. 1). Lakes, coastal and transitional waters were most often assessed by abundance metrics (Table 4). The ecological status of rivers was most often assessed by sensitivity and ecological trait metrics, followed by abundance metrics.



Fig. 2. Geographical distribution of the ratio of national methods resulting from this study compared to the number of methods required for national Water Framework Directive monitoring (country $\times$ water categories $\times$ biological quality elements). Colour codes: black: $>75\%$, dark grey: $50\text{--}75\%$, light grey: $<50\%$ (including Cyprus and Malta not depicted on the map), white: not covered by the survey.

Sensitivity metrics were also widely-used in lake and coastal assessment while the assessment of transitional waters relied more on the assessment of ecological traits. Richness metrics held a considerable share (7–13%) among all water categories. This share was generally smaller for diversity metrics, which were usually applied by invertebrate-based methods according to MCA results. MCA also revealed that non-native taxa were used mainly within fish-based

methods for rivers and lakes, and rarely in transitional waters. Metrics targeting individual condition were often associated with angiosperm methods, sensitivity metrics with phytobenthos methods, and abundance metrics with phytoplankton methods.

The number of metrics used per method differed significantly between BQEs (H-Test; $p < 0.05$). Fish assessment methods had the highest number of metrics (mean: 7), followed by benthic

Table 4
Percentage of metric types used in water categories. See text for type descriptions.

		Coastal waters	Lakes	Rivers	Transitional waters
Taxonomy-based	Richness	13.1	7.2	12.1	11.4
	Abundance	47.3	46.2	16.4	42.4
	Diversity	5.7	5.5	4.5	4.7
	Assemblage composition	0.4	1.0	0.5	3.0
Autecology-based	Sensitivity	20.8	25.7	37.3	10.2
	Ecological traits	5.3	11.0	25.7	21.6
	Individual condition	7.3	0	0.5	4.7
	Alien	0	3.4	3.0	0.8
Non-biotic		0	0	0	1.3

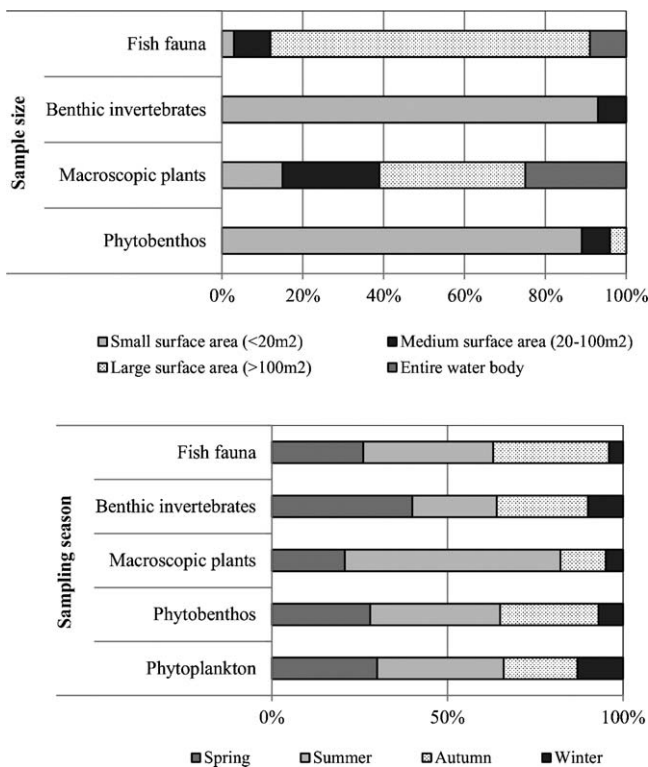


Fig. 3. Share of sample sizes and sampling seasons among biological quality elements.

invertebrates (4), macroscopic plants (4), phytoplankton (3) and phytobenthos (2). There were no differences between water categories (H-Test; $p = 0.15$).

3.5. Pressure-impact relationship

More than half of the methods appraised the impact of eutrophication or organic pollution (Fig. 1). This especially applied to methods evaluating lakes and coastal waters where almost 2/3 of methods focused on these stressors. Hydrological or morphological deterioration was detected by 27% of methods, most apparent for transitional waters. Other water quality aspects were addressed by 13%. More than half of the river methods focused on pressures other than eutrophication or organic pollution.

The share of methods detecting eutrophication or organic pollutions was markedly decreasing from autotrophic to heterotrophic elements: phytoplankton > phytobenthos > macroscopic plants > benthic invertebrates > fish fauna. An almost reverse order was revealed for hydrological or morphological deterioration: Fish fauna and macroscopic plants > benthic invertebrates > phytoplankton > phytobenthos. About 20% of the methods for each biological element (except phytoplankton) addressed other water quality aspects.

The pressure-impact relationship has not been tested or documented for one-third of the methods, with clear differences between water categories: For 46% of the methods applied to transitional waters the relationship was not documented, followed by methods for coastal waters (37%) and rivers (31%), while lake methods were tested in 81% of cases. Methods using fish fauna or macroscopic plants were checked least frequently (44% and 43% unchecked, respectively) while phytoplankton, phytobenthos and benthic invertebrates were the best validated biological elements (more than 75%). The number of case-studies to empirically validate the pressure-impact relationships was unequally distributed between water categories and biological elements

Table 5

Median coefficients of correlation and number of observations to validate the pressure-impact relationships per biological element (N = number of case-studies).

Biological element	Correlation coefficient	N	# Observations	N
Phytoplankton	0.76	30	400	34
Phytobenthos	0.56	18	202	17
Macroscopic plants	0.71	24	104	32
Benthic invertebrates	0.64	33	103	36
Fish fauna	0.55	7	484	12

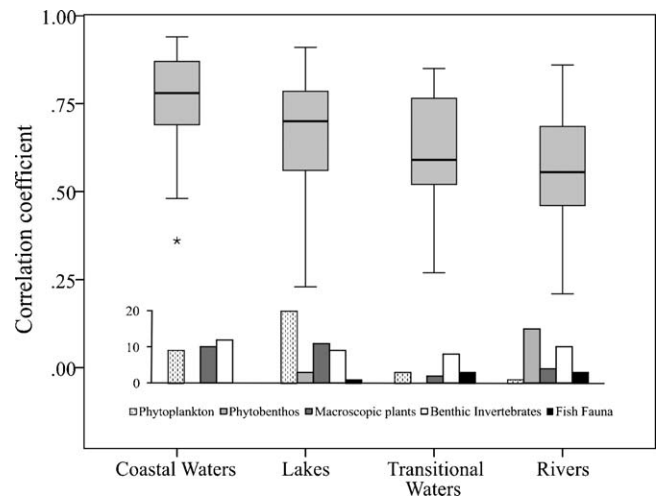


Fig. 4. Range of correlation coefficients gained from pressure-impact analysis in different water categories. The number of case-studies per water category is specified below (total number = 110).

(Table 5). Most studies were reported for lakes (especially phytoplankton), and least for transitional waters. Across water categories most case-studies referred to benthic invertebrates, and least to fish fauna. The majority of studies tested the response to gradients of nutrient enrichment or organic pollution. The strength of relationships differed significantly between biological elements and water categories (H-Test, $p < 0.05$). The correlation coefficients generally covered a broad range (<0.4 to >0.8), but on average with the pattern: phytoplankton > macroscopic plants > benthic invertebrates > phytobenthos and fish fauna (Table 5). In terms of water categories (Fig. 4) the following order resulted: coastal waters > lakes > transitional waters > rivers. The number of observations used to empirically validate the pressure-impact relationship differed between water categories and biological elements (Table 5). On average, relationships were established based on 250 samples (median value).

3.6. Quality class boundary setting

The class boundary setting of European assessment methods was mostly based on statistical principles (45%), i.e. use of statistical approach without any ecological approach, and 37% of assessment methods used ecological approaches alone or together with other approaches to define boundaries. In 18% of cases class boundary setting was limited to expert judgement (Fig. 1). For the development of lake assessment methods mostly ecological approaches were applied (58%), especially in the assessment of phytoplankton (82%) and benthic flora (60%). Statistics-based boundary setting dominated the classifications of rivers and transitional waters (56%). In these categories only 24% of methods were based on ecological boundary setting. This especially applied to benthic fauna and flora in rivers, and benthic flora in transitional waters, where an ecological approach was only used in about 20% of cases or less. In coastal waters the classification of phytoplankton and benthic flora

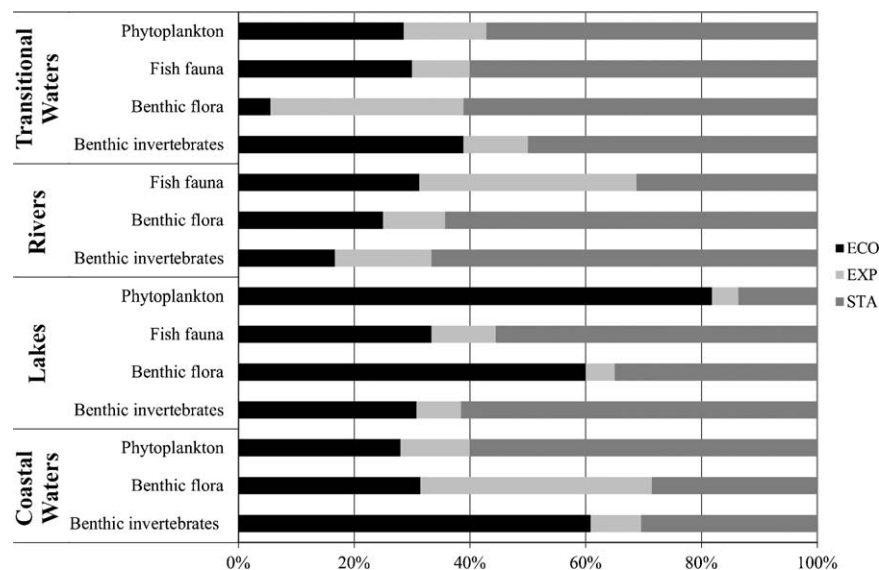


Fig. 5. Boundary setting approaches used in ecological assessment methods for water categories and biological quality elements. ECO-ecological; STA-statistical; EXP-expert judgement.

was mostly based on statistical definitions and expert judgement (approximately 70%) while the classes in invertebrate assessment were predominantly set based on ecological principles (61%).

Boundary setting approaches were considerably different between biological elements and water categories (Fig. 5). Overall, the highest share of ecological boundary setting applied to phytoplankton methods (47%) and the lowest to fish and benthic flora (31%). However, conspicuous differences were observed among water categories. The frequency within phytoplankton methods ranged from 82% for lakes to only 28% for coastal and transitional waters; within benthic flora methods values ranged from 60% for lakes to only 6% for transitional waters.

4. Discussion

Europe has decided to use the ecological status as the primary determinant of management needs for surface waters. The bioassessment methods reviewed in this paper act as principal indicators of this status. They form the link between the actual condition of the environment and its interpretation which, in turn, leads to management actions of either conservation or (often costly) restoration. With such a responsible role, aquatic bioassessment methods should ideally provide reliable outputs from an integrated appraisal of relevant structural and functional key variables of the ecosystem at low costs (e.g. Kurtz et al., 2001; Bonada et al., 2006). Since these all-in-one solutions are illusive, in practice aquatic bioassessment seems always to be a trade-off between the legislative requirements, best scientific knowledge and socio-political emphasis.

To characterise how this balance is accomplished in Europe we compiled essential data on bioassessment methods developed for the purpose of WFD monitoring. We knew from earlier publications that many such methods existed (e.g. Birk and Schmedtje, 2005; Borja et al., 2009; Poikane, 2009) and thus chose a survey design based on mail questionnaires. The ease of data accessibility and verification through pertinent networks (e.g. ECOSTAT and its expert groups) compensated for probable disadvantages of this design (e.g. ambiguous replies, refusal of information, lack of face-to-face communication; Galpin and James, 1984) and we are thus confident that our survey covers the vast majority of methods available.

At first glance the multitude of aquatic bioassessment methods used for European surface waters is perplexing. One is tempted

to query if this methodological patchwork allows for comparable status classification across the continent. A Europe-wide record of ecological status, and the practice of river basin management in particular, demand for harmonised assessment concepts and approaches. Comparability is currently addressed by an extensive intercalibration exercise (e.g. Birk and Hering, 2009; Borja et al., 2007). Our findings point out the generally demanding character of such an exercise regarding the high number of different methods.

But looking beyond intercalibration, the question arises if this methodological diversity is necessary or counterproductive for a successful WFD implementation. The outcomes of our research questions reveal distinct patterns among methods assessing the same quality elements or water categories. We thus discuss these findings individually with regard to their merits and shortcomings.

4.1. Coverage of aquatic bioassessment

The low availability of methods in Eastern and Southern Europe reflects the different monitoring traditions and the fact that in the early 1990s, even for rivers, only half of the European countries were assessing biological parameters in addition to their physicochemical monitoring (Hering et al., 2003). Furthermore, it is not surprising that there are few methods for transitional waters since only 16 countries have designated water bodies in this category. Additionally, lagoons and estuaries are particularly challenging water bodies for the development of ecological assessment methods due to their dynamic and naturally stressed character (Elliott and Quintino, 2007). Zaldívar et al. (2008) emphasised that the implementation of the WFD is particularly problematic in transitional waters and listed several reasons including the difficulty to distinguish between natural and man induced stress (see also Dauvin and Ruellet, 2009).

The high availability of benthic fauna methods reflects the considerable tradition (Rosenberg and Resh, 1993) of macrozoobenthos based aquatic biomonitoring due to the limited mobility, variety of traits and adaptations of benthic animals. The assessment of macroscopic plants seems also quite well established as they are mostly fixed and sensitive to biotic and abiotic changes (Orfanidis et al., 2001). However, the high number of methods with untested pressure-impact relationships points at a general shortcoming. It is also remarkable that phytoplankton assessment that builds on a long tradition still mostly relies on

chlorophyll-a concentration measurements instead of more comprehensive methods, partly due to high spatio-temporal variability and community complexity, partly due to cost intensive identification.

4.2. Sampling

Sampling is fundamental to aquatic bioassessment as it provides the base data on which water bodies are classified. An ideal strategy combines the aspects of high precision and representativeness to detect relevant changes in ecological status of the entire water body. Since sampling represents a considerable part of the costs of bioassessment it often represents a trade-off between these aspects. Sampling practices applied in Europe are manifold, but our outcomes reveal two main strategies related to the organism groups used in bioassessment. The sampling of small surfaces is used for benthic invertebrates and phytobenthos. Organisms of these groups are characterised by small body sizes, high taxonomic diversity, and their distribution is related to microhabitat-structures. Single samples can contain many and/or highly abundant taxa whose identification often requires laboratory treatment and specific expertise (Kelly et al., 1998; Bonada et al., 2006). Processing invertebrate or phytobenthos samples is time-consuming and demanding, thus only performed for limited surface areas (Friberg et al., 2006). The basic assumption of this strategy is that the sample reflects the condition of the whole water body.

Macroscopic plants and fish are surveyed at larger spatial scales. Both groups feature smaller numbers of taxa relevant in bioassessment, and their organisms are usually larger in size (i.e. detectable and identifiable by the “naked” eye). The plants are themselves important structural components of the ecosystem and can form vast stands (e.g. seagrass meadows). Such formations often contain few taxa, but their extension is of ecological relevance, hence requiring large-scale surveys (e.g. Juanes et al., 2008; García et al., 2009). The fish fauna shares similar integrative features that demand extended sample surfaces: Fish habitats are considerably large, and most fish taxa are highly mobile, thus indicating conditions at larger spatial scales (e.g. Schiemer, 2000). Here, spacious sampling, e.g. done by boat survey or aerial photography, prevails at the cost of a lower sample precision.

For effective management of the aquatic resources representative sampling is a fundamental, yet highly theoretical concept. Few studies exist that investigate necessary sampling area and efforts in relation to water body size, its spatial heterogeneity or temporal dynamics (e.g. Carstensen, 2007). As sampling strongly determines the results of status classification its strategy shall ideally be aligned to the individual biological indicator used in monitoring (de Jonge et al., 2006). But in practice many assessment methods were built on the basis of already existing data (Beliaeff and Pelletier, 2011).

4.3. Taxonomic resolution

According to the niche-concept in ecology each species is adapted to an individual suit of environmental conditions (Hutchinson, 1957; Devictor et al., 2010). Since anthropogenic pressures affect particular conditions, their impact is best indicated by monitoring at species-level (e.g. Resh and McElravy, 1993; Lenat and Resh, 2001; Lavoie et al., 2009). This level is used by most assessment methods reflecting the well-established taxonomy and good expertise generally existing across Europe. While species-level is commonly attainable for the less diverse groups of macroscopic plants and (freshwater) fish the benefits of identifying benthic invertebrates to higher taxonomic levels is a recurring theme in scientific literature (e.g. Stubauer and Moog, 2000; Dauvin et al., 2003; Bertasi et al., 2009). Besides sufficient bioindicative

capacity the supporters of coarse taxonomic categories advocate the reduced risk of wrong determination (and, hence, misclassification of ecological status), less expertise required and, ultimately, lower costs of biomonitoring (e.g. Pik et al., 1999; Mistri and Rossi, 2001). However, when investigating the response of multimetric methods to gradients of human pressure, methods using species-level could probably be more sensitive in evaluating the impacts than those using higher taxonomic levels, especially in multi-pressure environments (Borja et al., 2011). But for those methods that operate at class- or phylum-level, or without any reference to taxonomic information, we see room for improvement by considering data at a finer taxonomical scale (e.g. Tuvikene et al., 2010).

4.4. Biological metric selection

The metric types used differ considerably between BQEs and water categories. An important factor in the use of metrics seems to be monitoring tradition. For example, in lakes, the metric chlorophyll-a, a measure of algal abundance, is present in all assessment methods, resulting in a higher frequency of abundance metrics in lakes. For rivers, there is a long tradition in using benthic invertebrates for assessing the effects of organic matter (i.e. Saprobic Systems or ASPT, e.g. Birk and Schmedtje, 2005). Further method development has built to a large extent on this tradition, resulting in a relatively high frequency of sensitivity metrics for benthic invertebrates. For the BQE fish, assessment methods were developed more recently in the framework of scientific projects specifically targeting the WFD (Schmutz et al., 2007), resulting in a more balanced, multimetric approach.

The WFD defines ecological status as “an expression of the quality of the structure and functioning of aquatic ecosystems associated with surface waters” (WFD art. 2), but relies mostly on metrics for taxonomy and ecosystem structure to quantify it. A widely accepted interpretation of Article 2 is that metrics directly quantifying ecosystem functioning are not specifically required as it is implied that good structure alludes good functioning (Solimini et al., 2009), discouraging their development. One can argue that autecology-based methods are a proxy for real functional metrics – but this has not been properly tested yet (Sandin and Solimini, 2009).

Given the complex requirements of the WFD, it is not surprising that multimetric indices are common as BQE level assessment methods. The number of metrics per method is approximately three for all BQEs, except for fish where on average eight metrics are combined. There is no obvious ecological reason for this difference, and it is most likely caused by the difference in monitoring traditions discussed above. The choice of metrics has been driven to a large extent by the need to fulfil the very detailed requirements of the WFD, leading to multimetric indices covering the main aspects of the WFD Annex V – abundance, species composition, and autecology-related aspects. As a result, other relevant aspects, most notably ecosystem functioning, have been underrepresented.

4.5. Pressure-impact relationship

The main emphasis of Europe's aquatic bioassessment is still on the classical burdens of nutrient and organic pollution. This reflects the current situation, especially for receiving water bodies such as lakes and coastal waters: excessive nutrient inputs from diffuse sources continue to pose major threats for the European environment (EEA, 2010). Densely populated areas in the European lowlands and especially at estuaries suffer from direct human activities causing morphological alteration and habitat loss (Borja et al., 2006; CIS, 2006; López y Royo et al., 2009). Hence, these stressors are primarily addressed in rivers and transitional waters appraising medium- to large-scale habitat quality as indicated by

the fish fauna or macroscopic plants. The variety in pressure focus seems to mirror the multidimensional perturbation generally acting on European rivers.

The high number of methods with untested pressure-impact relationships is alarming. What do these methods actually assess? They often were developed based on theoretical concepts including a large deal of expert knowledge (e.g. Teixeira et al., 2010). Empirical testing was never conducted, justified by a lack of data or the general conditions unfavourable for bioassessment (see e.g. the Estuarine Quality Paradox; Elliott and Quintino, 2007). Our results seem to reveal a relationship between the degree of the system's natural variability (from transitional waters to lakes) and the percentage of methods untested. However, empirical validation of the pressure-impact relationship is indispensable for any method's development (Borja et al., 2011). Environmental management relies on bioassessment to detect anthropogenic impacts and their causes (Cairns, 2003). In return, the effects of measures to enhance the ecological status need to be indicated, given the high economical and socio-political relevance of aquatic resource management (Feld et al., 2011).

Our summary of validated pressure-impact relationships demonstrates that successful linkages of human disturbance and biological responses are possible. We are aware of the limited representativeness of these results owing to the possible bias introduced by the questionnaire replies and the enormous coefficient ranges caused by the heterogeneous conditions covered in the case-studies. Nevertheless, some general patterns clearly emerge: Similar to the findings discussed above, the best relationships are shown for methods of the naturally less disturbed ecosystems (deep lakes, coastal waters) and the classical indicators (phytoplankton). Correlation coefficients above 0.7, for instance, suggest that more than half of the biological variability is explained by pressure effects. The small datasets underlying most analyses highlight the preliminary character of these studies. In summary, outcomes are promising but more effort is needed in order to develop a comprehensive understanding of the human pressures detected by the individual methods. In particular there is a need to better understand cause (human pressure) - effect (metrics or indicators) relationships for highly integrative BQEs such as fish or plants. This is required to decide which management actions should be implemented regarding these BQEs.

4.6. Quality class boundary setting

Boundary setting is one of the most critical steps in the design of assessment methods as it defines the target values for environmental management. Common concepts promote the target of stable and healthy ecosystems (e.g. Karr and Chu, 1999). At the desired state, the long-term capacity to supply ecosystem goods and services is sustained (Rapport et al., 1998). Ecological thresholds, i.e. small changes in an environmental driver that trigger major changes in the ecosystem, should play a key role in boundary setting (Groffman et al., 2006; Lyche Solheim et al., 2008). Furthermore, target values shall reflect environmental conditions that are socially desirable or acceptable (Smyth et al., 2007).

But in practice boundary setting often follows non-ecological principles. Statistical approaches, for instance, in which the gradient of biological condition is divided into equidistant classes (Karr and Chu, 1999; Erba et al., 2009) allow for a convenient mapping of the ecosystem status but lack biological significance (Kelly et al., 2007; Brenden et al., 2008; Grenier et al., 2010). Prerequisites for the definition of ecologically meaningful boundaries are well-established pressure-impact relationships (Karr and Chu, 1999; Davies and Jackson, 2006). However, as outlined above, these do often not exist due to limited data availability or significant knowledge gaps. Though not entirely clear from the analysed data,

we assume that many methods have established pressure-impact relationships, but nevertheless use statistical approaches to define class boundaries. This reflects the fact that ecological thresholds or tipping points are an important ecological concept, but rarely detectable in a multi-pressure environment (Groffman et al., 2006). If pressure-impact relationships have not been established, expert judgement remained the last resort in national boundary setting. Moss et al. (2003) and Moss (2008) even argue that boundaries should generally be based on expert opinion since degrees of ecological status are not absolute entities but matters of judgement. This is true at least from a philosophical perspective as normative statements generally are beyond the descriptive character of science (Valsangiacomo, 1998).

We recommend more efforts to include ecology-based elements in the boundary setting of all methods, at least to define pressure-impact relationships. This comprises identifying crossovers of antagonistic structural or functional community features, e.g. the increase in phytoplankton biomass leading to secondary effects on the macrophytes (European Commission, 2010; Kuhar et al., 2010), or investigating the existence of ecological thresholds, e.g. indicated by non-linear pressure-impact relationships (Lyche Solheim et al., 2008). The selection of biological metrics determines, to a large degree, whether such phenomena are detectable. Non-linearities in the pressure response are most likely picked up by single metrics sensitive to a specific pressure, but may remain undetected by a multimetric index sensitive to multiple pressures. Also the choice of appropriate tools can support the detection of ecological thresholds. However, there is an apparent gap between the prominence of present theoretical frameworks involving ecological thresholds and regime shifts, and the paucity of efforts to conduct statistical tests on the actual appearance of such phenomena in ecological data. A wide range of statistical methods and analytical techniques have been proposed (e.g. Denoel and Ficetola, 2007; King and Richardson, 2003; Muggeo, 2003; Qian et al., 2003; Loughheed et al., 2007; Grenier et al., 2010). Yet, their application has been sparse and disturbance thresholds are mostly identified through visual inspection of data series plots (Lyche Solheim et al., 2008; Penning et al., 2008; Marchetto et al., 2009; McFarland et al., 2010), reflecting the fact that water managers demand for simple and transparent methods.

Since most classifications rely on non-ecological principles there is no guarantee that ecological class boundaries correspond to meaningful changes in ecosystem functions and biological communities. This results in European ecological targets being built on the basis of statistical distributions rather than ecological impacts. Therefore, we advocate an enhancement of national boundary setting rationales by implementing ecological components into quality classification.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.ecolind.2011.10.009](https://doi.org/10.1016/j.ecolind.2011.10.009).

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